

Samyuktha Kapoor

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SUMMARY

Data Analyst and AI researcher with expertise in developing machine learning models and automated forecasting pipelines. Skilled in Python, TensorFlow, statistical modeling, and translating complex requirements into scalable solutions in energy and healthcare domains.

EDUCATION

Northeastern University

Sep 2023 - Dec 2025

Master of Science, Data Analytics Engineering (GPA: 3.8)

- **Achievements:** Teaching Assistant, IE 7275 - Data Mining in Engineering
- **Coursework:** Database Management Analytics, Data Warehousing, ML Operations, Product Management, Computation & Visualization

Anna University

Aug 2019 - Apr 2023

Bachelor of Engineering, AI & Biomedical Engineering (GPA: 3.7)

- **Coursework:** Data Structures & Algorithms, Microprocessors, Digital Signal Processing, Probability & Statistics, Mathematics

EXPERIENCE

Massachusetts Clean Energy Center | *Data & Business Intelligence Analyst*

Jan 2025 - Sep 2025

- Built 12 automated Power BI dashboards tracking operational metrics across 5 departments on cloud platforms, reducing manual reporting time from 20 hours to 12 hours weekly and enabling real-time portfolio monitoring for \$50M+ in clean energy investments
- Engineered reusable DAX calculation framework processing 500K+ rows of grant data, eliminating Power Query bottlenecks that previously caused 2-hour refresh delays and enabling same-day decision support for executive leadership
- Collaborated with MassMappers GIS team to build ArcGIS dashboard mapping \$3.8M in clean energy investments across Massachusetts, enabling policymakers to identify 8 underserved districts and prioritize \$5M in equity-focused funding
- Automated SQL-based data quality checks across CRM and Power BI platforms, detecting 150+ compliance violations and reducing data accuracy errors from 12% to 8% through daily reconciliation workflows and anomaly detection

Northeastern University | *Data Science Research Assistant*

Aug 2025 - Present

- Developing AI diagnostic system using deep learning quality-aware attention networks to analyze Visual Evoked Potential signals, improving baseline classification accuracy from 91% to 94% across 2,500+ patient EEG recordings
- Built containerized ML deployment pipeline using Docker, Git, and Python (scikit-learn, TensorFlow) to preprocess clinical EEG data, train diagnostic models, and validate pre-conscious brain signal patterns with 99.2% data integrity across experiments and implemented model deployment using MLflow

Rudra Enterprises | *Business Data Analyst*

Aug 2022 - Aug 2023

- Analyzed performance metrics across 85 product lines using SQL queries on Oracle Systems, identifying 12 underperforming SKUs that led to 38% reduction in stockouts and \$250K in recovered inventory costs over 6 months
- Built customer segmentation models using Python (Pandas, scikit-learn) analyzing purchase behavior across 50K+ transactions, enabling targeted retention campaigns that improved repeat customer rate from 31% to 38%
- Created executive dashboards in Excel and Tableau visualizing sales trends, inventory turnover, and customer lifetime value metrics across 15+ regional markets, informing product positioning strategy, demand forecasting, and time series analysis

Indian Institute of Technology | *Machine Learning Intern*

Jul 2021 - Sep 2021

- Preprocessed and standardized breast cancer histopathology image dataset (10,000+ samples) using Pandas, Python statistical libraries, and Matlab, enabling CNN classification model achieving 97% validation accuracy, research published in IRO Journal

ACADEMIC PROJECTS

Audio Intelligence Platform | [Link](#)

- Engineered end-to-end MLOps pipeline using Apache Airflow, Docker, and MLflow to process 500+ hours of podcast audio, implementing parallel DAG execution and automated quality validation, achieving 99% data integrity

Payroll Data Warehouse & Analytics System | [Link](#)

- Built enterprise data warehouse integrating 4 HR data sources via Snowflake ETL pipelines into PostgreSQL Star Schema optimized for sub-second query performance, processing 100K+ employee records across 5 years of payroll history

YouTube Performance Analysis Pipeline | [Link](#)

- Engineered scalable ETL pipeline on AWS (S3, Lambda, Glue, Athena) with automated data transformations processing 500+ videos and 10M+ engagement metrics, implementing Tableau visualizations that identified content strategies increasing average view duration by 18%

SKILLS

- **Programming & Analytics:** Python (Pandas, NumPy, Scikit-Learn, TensorFlow), R, SQL (MS SQL, PostgreSQL), Statistical Analysis
- **Business Intelligence:** Power BI, Tableau, Looker, Excel, DAX, Power Query, Data Modeling, Data Visualization
- **Data Engineering:** Apache Airflow, Docker, Git, ETL Processes, Data Warehousing, AWS (S3, Lambda, Glue, Athena), GCP
- **Databases & Tools:** PostgreSQL, MongoDB, Snowflake, Oracle Systems, Talend, APIs, MLflow, Jupyter Notebook, VS Code
- **Machine Learning:** Classification, Random Forests, Feature Engineering, Forecasting, Time Series Analysis, Deep Learning